Audits for Bias in AI Systems

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WHEN AI GOES WRONG

Incidentdatabase.ai

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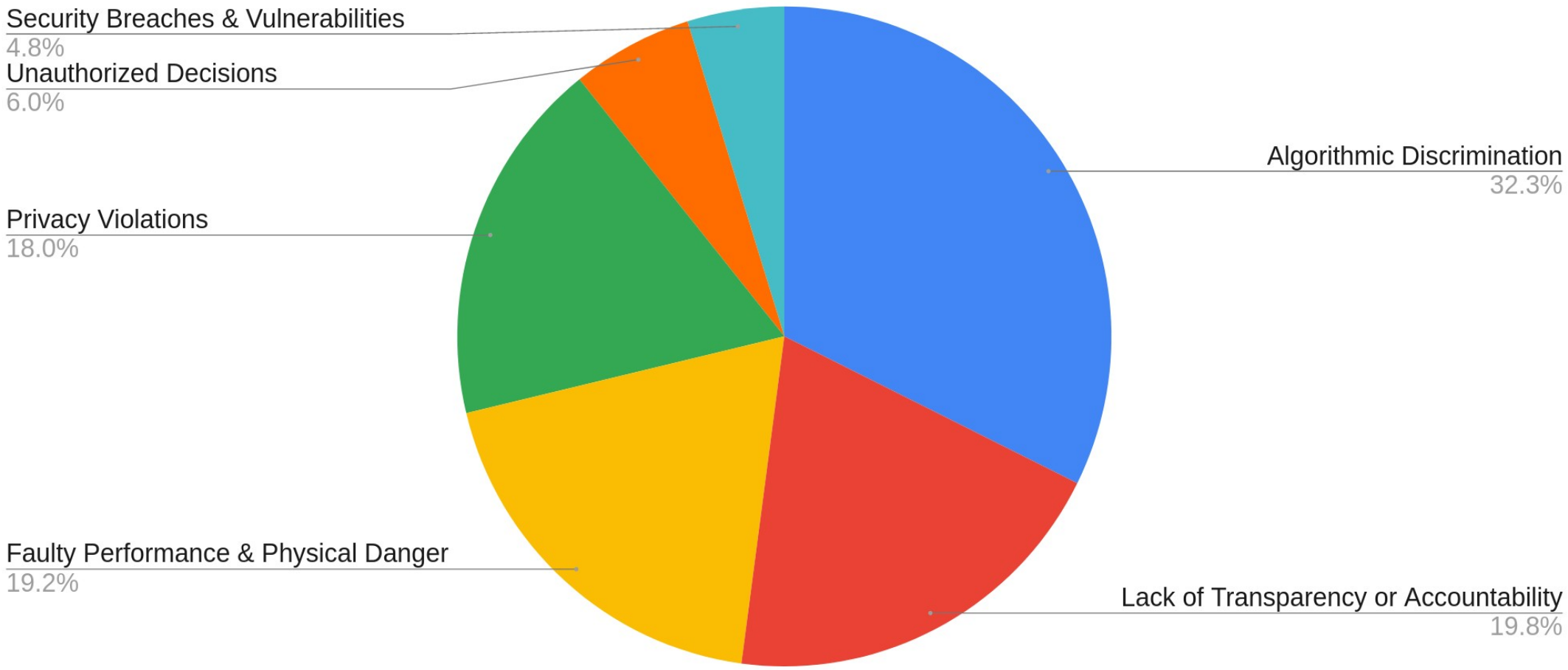
COMMON FAILURE MODES



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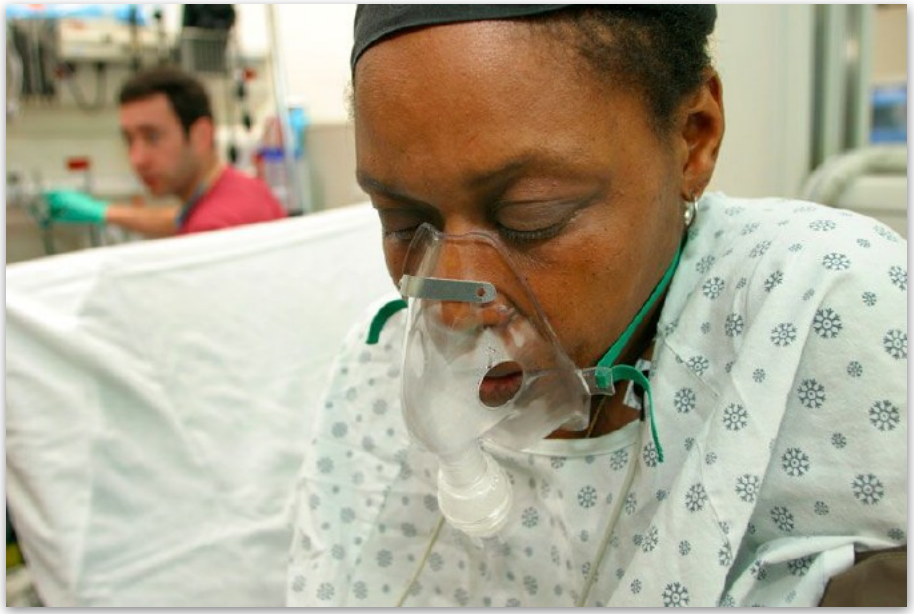
Harms Are Often Concentrated on Marginalized Groups

- Lack of access
  - Screenout
  - Digital Divide

- Bias and opacity in:
  - Credit scoring and credit products
  - Criminal risk assessment instruments (RIAs)
  - “Ed tech”
  - Employment screening systems
  - Fraud detection systems
  - Healthcare resource allocation
  - “Landlord tech”
  - Predictive policing systems



Millions of black people affected by racial bias in health-care algorithms

- Nature, 24 Oct. 2019

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Fairness: Discrimination Testing

Testing for Discrimination:

- Group disparities
  (preferably using tests with legal standing)

- Individual disparities
  - Counterfactuals
  - Track decision boundaries
  - Train adversary models or use special training constraints

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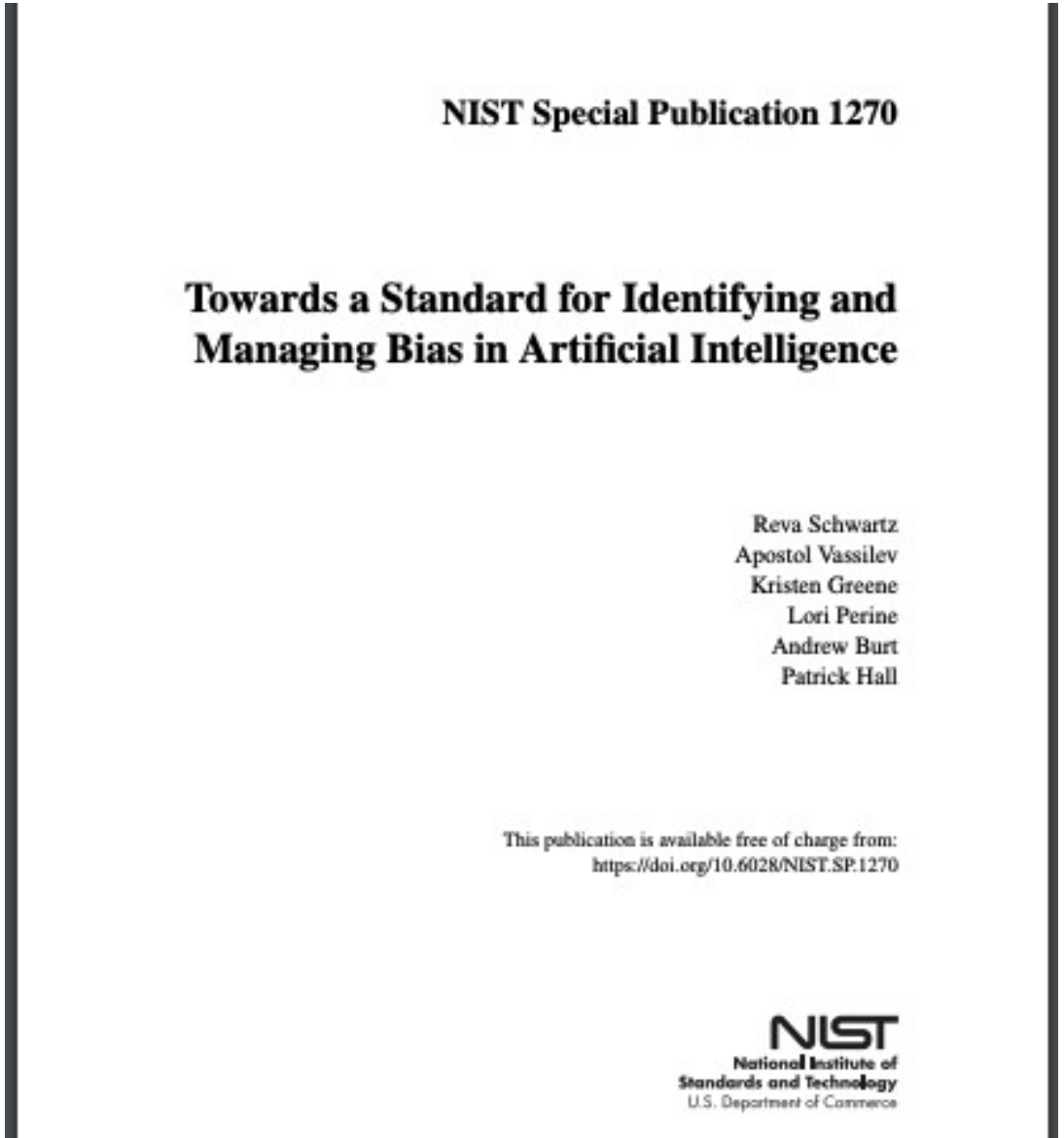

NIST Special Publication 1270

Towards a Standard for Identifying and Managing Bias in Artificial Intelligence

Reva Schwartz
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NIST
National Institute of Standards and Technology
U.S. Department of Commerce

AI AUDITS

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What is an Audit?

- **Official tracking of adherence to policy, regulation, or law**
  - Legal standards or regulatory frameworks
- **Conducted by independent third parties**
  - Internal, external
  - Transparent and fair
- **Reporting outcomes based on audience and purpose; accountability**
  - Public; regulators; internal overseers
  - Agreed upon standards
  - Certifications
  -

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Measuring Bias

More than data and model outputs

- Unconscious bias and overt prejudice
- Initial design choices (“screenout”)
- Homogenous engineers/perspectives
- Socio-technical applications

Measurement Challenges

- Data labeling
- Demographic descriptions (BISG)
- Language limitations
- Benchmarks
-

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Tangible Practices

- **Mathematical definitions of bias**
  - Avoid overly complex, sophisticated or unexplainable thresholds
- **Convert outcomes to a binary or single numeric outcome**
  - Apply traditional measures of practical and statistical significance
  - Select appropriate statistical measures (AIR, *t*-tests, Fisher's exact test)
  - Align traditional metrics to existing laws and regulations

Example: Bias Audit Methodology

- Score each video with FakeFinder models.
- Segment scores by demographic group
  (intersectional groups not considered in this case).
- Establish protected groups:
  East Asian, Black, South Asian, and Women.
- Establish control groups: Whites and Men.
- Test for practical and statistical significance in ***outcome*** differences:
  - **Statistical significance:** *t*-test, significance at $p = 0.05$
  - **Practical significance:** adverse impact ratio (AIR)
- Acceptable threshold: 0.8 – 1.25 (4/5th's rule)
- Ideal threshold: 0.9 – 1.11
- Test for practical significance in ***performance*** differences:
  - **Practical significance:** Accuracy, TP, TN, FP, FN rates
- Acceptable threshold: 0.8 – 1.25 (4/5th's rule)
- Ideal threshold: 0.9 – 1.11

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Example results

- **Practical Significance: AIR**

For every 1000 deepfakes detected with White faces, we expect 694 deepfakes with S. Asian faces to be detected.

Demographic Groups	AIR
E. Asian-to-White	1.004
Black-to-White	0.821
S. Asian-to-White	0.694
Female-to-Male	1.035

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Example results

- **Statistical Significance: t-Tests**

True positive scores for White faces are on average 2.53% lower than for S. Asian faces. This difference is significant, but the actual difference is moderately small. Sample size and a narrow standard deviation for S. Asian scores contribute to the statistical significance, but so does the difference in group means.

Demographic Groups	Control Mean	Comparison Mean	Percent Difference	*p*-value
E. Asian-to-White	0.948	0.964	-1.69	3.39E-04
Black-to-White	0.948	0.926	2.32	6.65E-02
S. Asian-to-White	0.948	0.972	-2.53	4.06E-04
Female-to-Male	0.955	0.948	0.73	1.62E-01

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Example results

- **Performance and Error Rates**

E. Asian faces experience 644% of the false positive rate that White faces experience.

Demographic Groups	Acc. Ratio	TPR Ratio	FPR Ratio	TNR Ratio	FNR Ratio
E. Asian-to-White	1.005	1.012	6.438	0.973	0.394
Black-to-White	0.969	0.951	0.000	1.005	3.488
S. Asian-to-White	1.017	1.020	0.000	1.005	0.000
Female-to-Male	0.988	0.987	#DIV/0!	0.992	2.276

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Example: Audit Conclusions

- **Do deep fake (detectors) discriminate?** Yes, like nearly all other socio-technical AI systems.
- **Bias tests indicate disparity in both outcomes and performance.** Performance ratios point to problems in erroneous decisions. (High-confidence erroneous decisions are a common failure of neural networks.)
- **Biased and wrong deep fake detection consequences.** Bias causes wrong decisions and allows for adversarial exploitation.
- **Remediation via technical or process means is essential.**
- **Analysis via causal or explainable AI (XAI) methods is required to understand drivers of bias.**

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NYC Local Law 144

- **Requirements:** Companies using an automated decision tool in their employment process provide notice, audit the system, and publish results
  - Employers or employment agencies who use such tools must use an external auditor
  - Output tables are provided, in alignment with existing EEOC practices
- **Areas of Contention:** Definitions, required data, relationship between vendor providers and complying employers
- **Milestones:** First formal “AI Audit” law

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RISK & COMPLIANCE JOURNAL

New York’s Landmark AI Bias Law Prompts Uncertainty

Companies that use AI in hiring are trying to determine how to comply with a New York law that mandates they test their systems for potential biases

LOCAL LAWS
OF
THE CITY OF NEW YORK
FOR THE YEAR 2021

No. 144

Introduced by Council Members Cumbo, Ampry-Samuel, Rosenthal, Cornegy, Kallos, Adams, Louis, Chin, Cabrera, Rose, Gibson, Brannan, Rivera, Levine, Ayala, Miller, Levin and Barron. Passed under a Message of Necessity from the Mayor.

A LOCAL LAW

To amend the administrative code of the city of New York, in relation to automated employment decision tools

Be it enacted by the Council as follows:

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COMPLEXITIES

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CONTEXTUAL BIAS

Fairness - what does it mean?

- **Algorithmic Fairness (1) -** Equally accurate for members of legally protected groups (statistical bias)
- **Algorithmic Fairness (2) -** The algo produces the same percentage of false positives or false negatives for each group
- Impossible to maximize both/achieve parity on all axes

Fairness - Conceptual; normative; legal

- **Conceptual -** A score has same meaning across groups - equal predictive accuracy. Not the best measure?
- **Normative -** Parity in ratio of FP/FN is normatively significant, indicative of likely unfairness
- **Legal:** The collection and use of data on protected classes is necessary for testing across either standard.

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TECHNOLOGY IN CONTEXT

- **Bias in Social Process**
  - Randomization
  - Predefined constraints
- **Design and Practice**
  - Watch for discriminatory outcomes in actual practice
  - Connect to context and policy controls (design of emergency services; policing routes;

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REGULATORY APPROACHES

- **Risk Assessment**
  - Future oriented
  - Take risks, and avoid harms
- **Policy Baggage**
  - Threshold question - Normative choice to use technology
  - Autonomy, dignity, privacy, equality, and human rights
- **EU AI Act**
  - Unacceptable risk → Prohibited Use
  - Values based, not technology effectiveness

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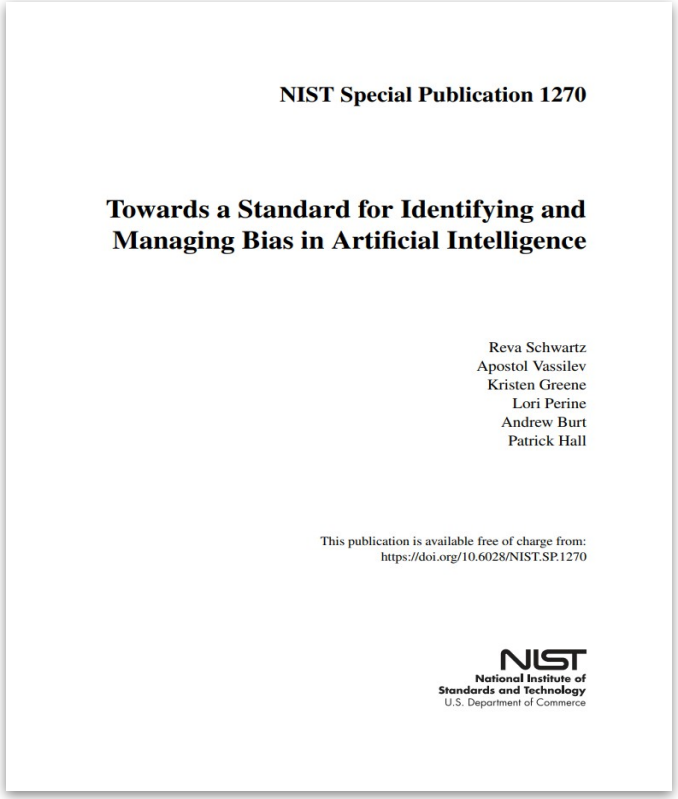
NIST - Governance Practices

NIST SP 1270 Bias Guidance



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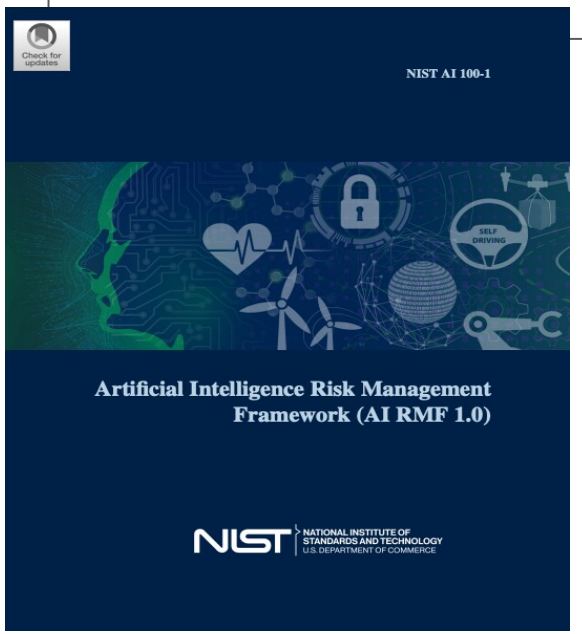
NIST AI Risk Management Framework

Core RMF: high-level perspectives and guidance

RMF Playbook: lower-level guidance and resources.



Check for updates

NIST AI 100-1

Artificial Intelligence Risk Management Framework (AI RMF 1.0)

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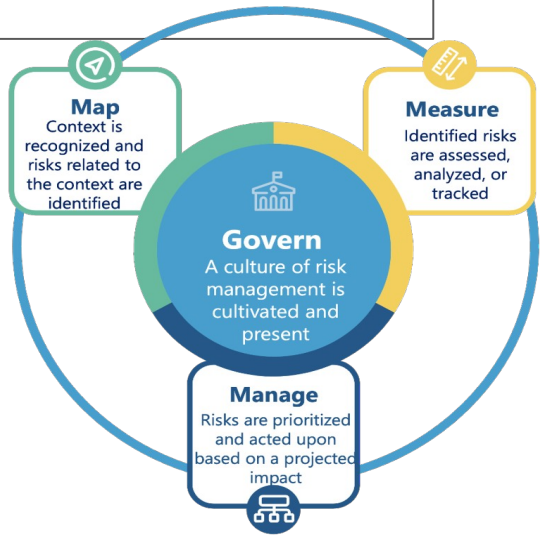

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Model Audits - Summary

- **Audits may have different meanings in the context of AI**
- **To what standard do we audit?**
  - Legal and industry standards
  - Who audits the auditors?
- **Audits are being incorporated** into regulatory and legal oversight systems

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THANK YOU -

QUESTIONS?

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